HEC HIGHER EDUCATION DATA WAREHOUSE (HEDW)

Project Proposal and Product Requirements Document

Prepared in the format of the HEC TDF proposal: Project Summary, Objectives, Introduction, Methodology, Development Plan, Impact and SDGs

PROPOSED TO	Higher Education Commission (HEC) Pakistan
PROJECT TITLE	Framework and Prototype for a Central Higher Education Data Warehouse
APPLICANT / PI	[To be completed by applicant]
COLLABORATORS	[To be completed: technology partner, pilot universities]
VERSION / STATUS	1.0 - Draft proposal for review
CORE PLATFORM	PostgreSQL + Python/SQL ETL + Power BI
NEW IN THIS VERSION	Use Case dimension and University Ideology dimension, both with full drill-down

Items shown in [square brackets] are applicant-specific and must be completed before submission. Timelines, targets and KPI definitions are proposals that require validation with HEC and the participating institutions. This document does not claim that HEC has approved or adopted the system.

PROJECT DETAILS

1. PROJECT SUMMARY

The aim of this proposal is to present a framework and a working prototype for a central Higher Education Data Warehouse (HEDW) that brings together the data of public and private universities and degree-awarding institutions (DAIs) of Pakistan for analytics and decision making. A central warehouse preserves historical information and improves the quality of education planning by tracking institutions and their performance: the number of enrolled students in each degree program, institutional capacity, faculty availability and retention, admissions and intake, graduation, research, scholarships and the accreditation process.

Priority Area: Universities across the world use data warehouses to support academic and policy decisions. For Pakistan, a unified analytical platform is needed so that national, provincial and institutional views of higher education are consistent and comparable. This proposal sets out a standard data format, a conformed dimensional model, a governed ETL pipeline and a dashboard layer for HEC. It does not assume which systems HEC already operates; any such statement requires validation with HEC.

Impact: A central HEDW lets analysts gather data from different institutions, prepare it for analysis, store it in a governed storage layer and consume it through analytics and visualization tools. Two analytical dimensions are added to the standard design: a **Use Case dimension**, which ties every KPI to the decision it supports, and a **University Ideology dimension**, which classifies institutions by mission and orientation so that like can be compared with like. Both support full drill-down.

Please also specify / state / elaborate briefly (1 sentence) the following:

- **Proposed idea funded from:** Support is sought from HEC Pakistan to carry the work from prototype to pilot [funding programme to be confirmed].
- **Lab scale / prototype:** A lab-scale prototype will implement the star schema in PostgreSQL, a Python/SQL ETL pipeline with a quality gate, synthetic source data and Power BI dashboards.
- **MS or PhD student work / title:** [To be completed by applicant].
- **Patent:** [To be completed by applicant].
- **Publications / Conference:** [To be completed by applicant].
- **Industrial problem or initiative:** The core problems can be divided into structural (fragmented, non-standard institutional data), organizational (no shared definitions, owners or submission calendar) and academic (limited ability to compare institutions, track cohorts and align programs with national needs).
- **ROI (Return on Investment):** A data warehouse is an investment in decision-support infrastructure and its value is not only financial. There is no single standard method for data-warehouse ROI; the key factors are net present value, internal rate of return, average rate of return over a time horizon and payback period. These will be estimated during the pilot, for example from reporting effort saved and faster policy turnaround.
- **Targeted sector of economy:** Academia, government bodies including HEC and provincial higher-education departments, and industry (through employability and skills-demand analytics).

2. PROPOSED GOALS / OBJECTIVES (please identify quantifiable goals)

Please clearly align the objectives with the output in the form of a **product** or **process**.

1. Determine the HEDW requirements together with HEC stakeholders: stakeholder register, business questions and a KPI catalogue of about 26 KPIs across 16 use cases (Steps 1 and 4).
2. Define a standard data-submission format for student, faculty, program, admissions, graduation, research, scholarship and finance data, with Excel/CSV templates and database-connector specifications.
3. Construct a conformed dimensional model of 13 dimensions, including the Use Case, KPI and University Ideology dimensions, and 9 fact tables, each with defined grain, keys and history-tracking policy.
4. Build the PostgreSQL warehouse and an ETL/ELT pipeline with raw, staging and quality-gate stages; every load produces an auditable quality report covering completeness, accuracy, consistency, uniqueness, validity and timeliness.
5. Run a pilot with at least 3 institutions of different types (public, private and DAI) using approved, non-sensitive or synthetic data.
6. Deliver a Power BI semantic model with national, provincial and institutional dashboards covering the pilot use cases, with drill-down by ideology group and use case.
7. Implement the security and governance framework: roles, row-level access, masking, audit logging, data ownership and stewardship.

8. Hand over source code, documentation and all rights to HEC Pakistan, together with a roadmap for advanced analytics and AI.

Objective	Output	Type
1	Requirements specification and KPI catalogue	Process
2	Data standard and submission templates	Process
3	Dimensional model and data dictionary	Product
4	Warehouse database and ETL pipeline	Product
5	Pilot report with quality scorecards	Process
6	Dashboards and semantic model	Product
7	Security and governance framework	Process
8	Hand-over package and AI roadmap	Product

3. INTRODUCTION (not to exceed one page)

The large amount of data generated by universities provides an excellent opportunity to derive insights for decision making [1]. A central HEC data warehouse would provide a general view of institutions across Pakistan and serve as a significant source of data for decision support and policy making [2]. Deploying an HEDW architecture opens opportunities for institutions to improve academic operations and services [4]. Such a warehouse also provides the foundation for the following:

- **Better data quality:** An HEDW centralizes data from transactional systems, operational databases and flat files [5]. It cleanses the data, removes duplicates and standardizes it so that there is a single source of truth.
- **Faster insights:** Data from many institutions can be integrated, letting decision makers set strategy with confidence [6]. HEC and institutions can plan for enrollment by degree program, institutional capacity, faculty retention, admissions or intake, required faculty for programs, grants and the accreditation process.
- **Smarter decision making:** The ultimate goal of an HEDW is to support large-scale academic intelligence, such as data mining to find unseen patterns, and machine learning [7]. Government and HEC representatives can use this to base decisions on evidence.

The HEC data warehouse is a flow process that gathers structured and, where needed, unstructured data from many universities into one repository. With all the data in one place, it is easier to perform analysis and reporting at different levels of combination. An HEDW consists of extraction, transformation and loading (ETL/ELT), an online analytical processing (OLAP) or semantic layer, consumer analysis tools and applications that manage data and deliver it to the business. The data warehouse life-cycle [8] describes the steps the system goes through while it is built. Figure 1 depicts this life-cycle.

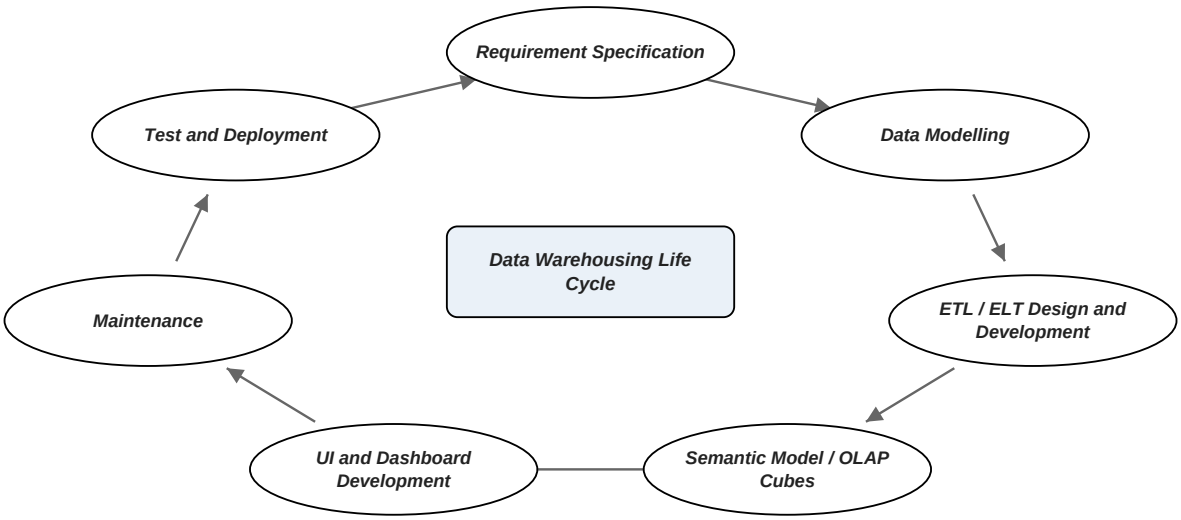


Figure 1: Data Warehouse Lifecycle, UI denotes the User Interface.

4A. BACKGROUND AND METHODOLOGY OF THE PROPOSED RESEARCH

Background: Deploying a robust HEDW architecture opens opportunities for universities [9, 11] to improve their services and operations, and for HEC to derive actionable analytics [10]. Public and private institutions differ in admission policies, facilities and research capacity, so aligning policies and standards requires data that is comparable across the sector. Institutional systems are typically transactional (OLTP) systems built for daily operations such as admissions, registration and payroll. They are not designed for cross-institution analysis or for answering questions that span many years. A multidimensional model with an OLAP/semantic layer gives HEC a sound decision-support system for such questions.

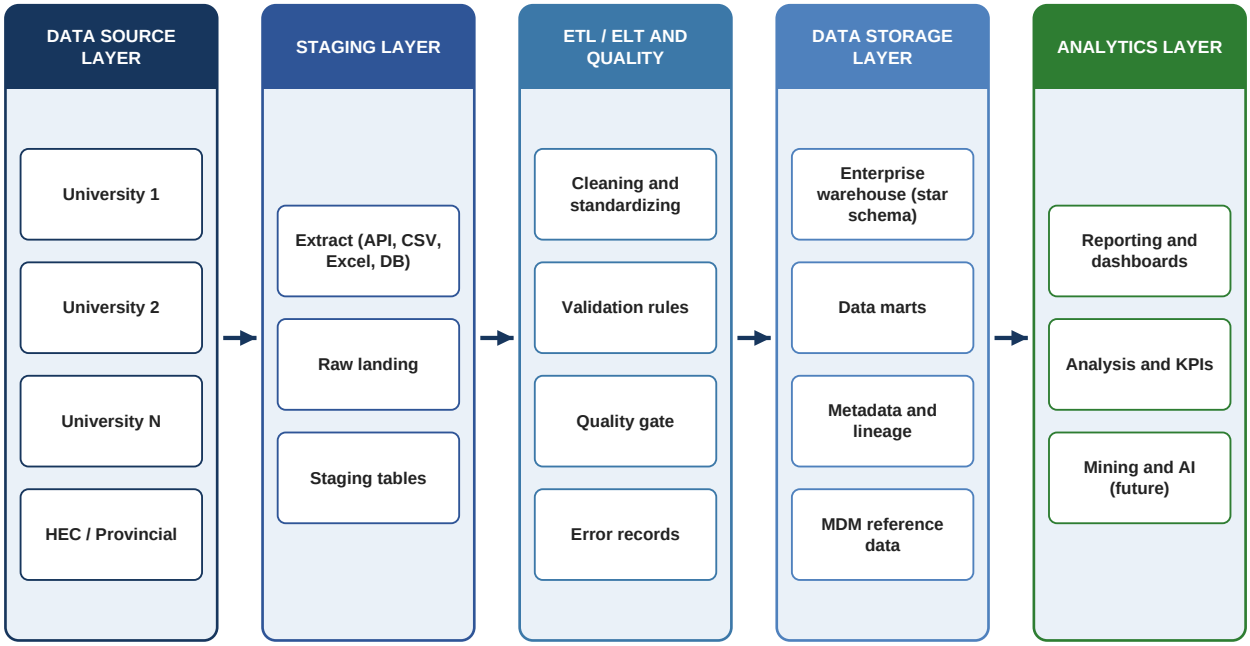


Figure 2: Architecture of the data warehouse for HEC Pakistan.

Methodology: The proposed HEDW has four main parts: data sources, a staging area, data storage and information delivery. The schema used throughout is a **star schema** with conformed dimensions shared across subject areas. HEC can recommend a standard data format to all universities and DAIs, to be submitted at the start or end of each year or semester. The main attributes to be collected are listed below. Direct identifiers (name, CNIC, passport, phone, email, address) are not needed for analytics; they are pseudonymized or excluded at load time under the security design.

Student data (currently enrolled or newly admitted): student ID / roll number (hashed), gender, birth year, nationality group, domicile district and province, admission date, program, prior-qualification board and marks (SSC / HSSC), admission category, student status.

Faculty data: faculty ID (hashed), gender, nationality group, institution and campus, date hired, designation / position held, promotion date, qualification, qualification institution and country, qualification year, area of specialization, employee type (permanent / visiting), total credit hours delivered. Salary data, if collected, is held only in aggregated form.

Alumni and academic data (start of each semester): alumni ID (hashed), institute, year of enrollment, year of graduation, CGPA obtained, specialization, organization sector, position and years of experience; number of students enrolled and courses offered per semester.

Institution and program data: institution identity and HEC/reference code, type, sector, status, location, ideology classification, program list, intake capacity and accreditation status.

These fields are provided by universities as Excel sheets or database files each year before the start of a new semester, which lets HEC maintain information about all students across Pakistan on one platform. Because most institutions keep data in their own database systems, Figure 2 shows the four main layers of the university data warehouse:

1. **Data Source Layer:** internal and external sources such as ERP systems, learning management systems and internal databases.
2. **Staging Layer:** temporary storage for data extracted from several sources during extract, transform and load processing.
3. **Storage Layer:** unified storage in a central location, on-premises or in the cloud. It can also take the form of data marts: subsets of the warehouse for departments such as student affairs or individual schools.
4. **Analytics Layer:** BI tools query the storage layer and share insights through reports, trends, visualizations, graphs and charts.

Extended dimensional model (new): To the standard dimensions (date, institution, student, program, faculty, province, district, discipline, gender, institution type) this proposal adds three: **Dim_Use_Case** and its child **Dim_KPI**, which tie every KPI to the decision and the user it serves, and **Dim_University_Ideology**, which classifies institutions by mission and orientation. Both new hierarchies support drill-down from national level to institution, program or KPI. Full specifications are in Annexes A, B and C.

4B. TECH DEVELOPMENT PLAN: SCHEDULE / PHASING (preferably with a time / Gantt chart, not to exceed one page)

Building an HEDW from scratch is the usual procedure we intend to follow. The project time frames in Figure 3 are approximate, because the duration of HEDW development depends on the collection of data, the complexity and quality of data in source systems, data security requirements and the analytics objectives. Close cooperation with universities, industry and a solution architect while defining the core and advanced functionalities helps avoid over-complicating the architecture and helps select the most cost-effective technology stack.

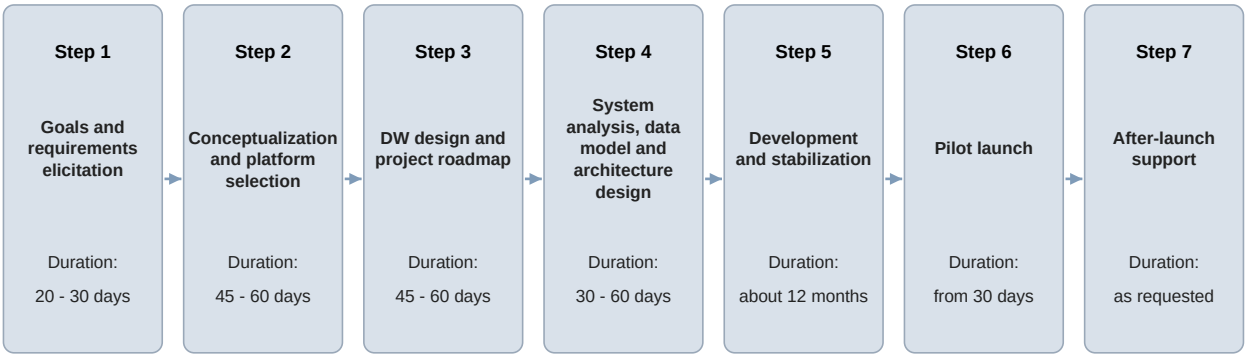
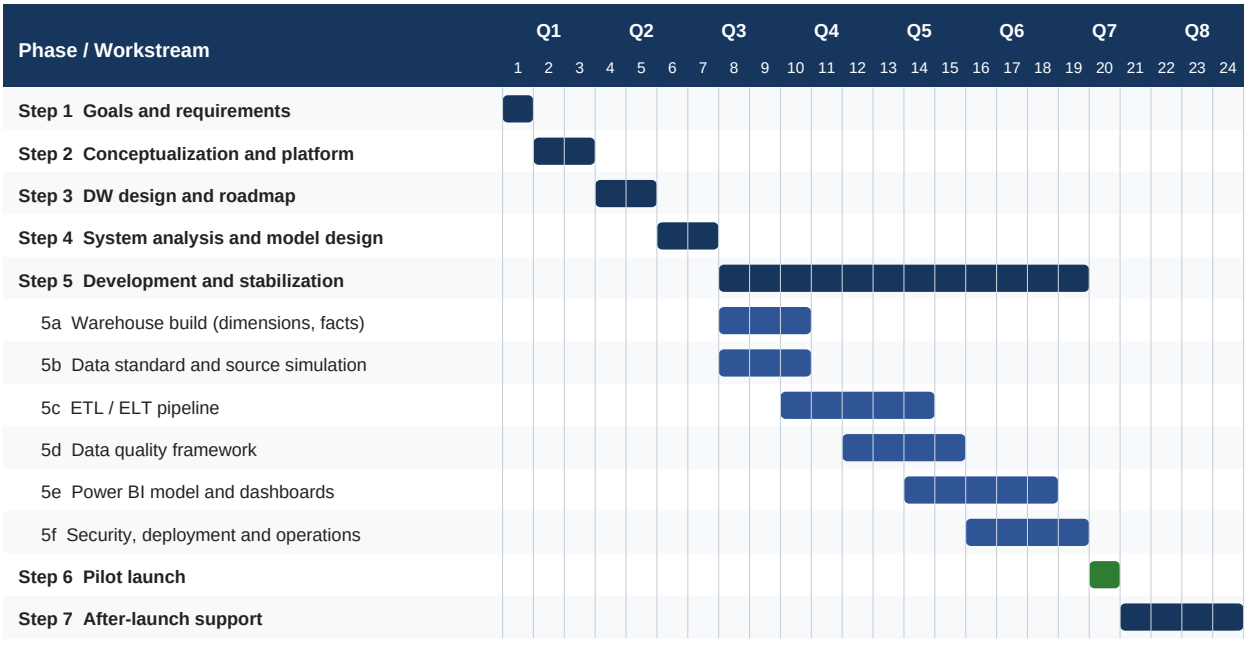


Figure 3: Time frames at various stages of the development of the HEC Data Warehouse.



Months from project start. Durations are indicative and depend on data availability and institutional participation.

Figure 4: Indicative Gantt chart (24 months including pilot and support).

4C. ACADEMIA-INDUSTRY INTERACTION PLAN: SCHEDULE / PHASING (preferably with a time / Gantt chart, not to exceed one page)

It should cover the schedule of meetings, visits, product testing, technology transfer, non-disclosure, material transfer and license signing (NDA, MTA and License signing).

The technical team of the proposed project will remain in close coordination with the pilot universities, HEC data owners and a technology partner [name to be confirmed]. Regular meetings will discuss the conceptual design, implementation, testing and deployment. Software solutions will be developed independently and jointly as agreed. All rights will be handed over to HEC Pakistan on completion of the project, after which HEC can decide its technical collaborator for continuous development and maintenance.

Interaction	Participants	Schedule
Kick-off and requirements workshop	HEC, pilot universities, project team	Month 1
Technical review meetings	Project team, technology partner, data stewards	Monthly
Steering committee	HEC sponsor, university representatives	Quarterly
Data-sharing agreement, NDA and MTA signing	HEC, each pilot university, technology partner	Before data onboarding (Month 7)
User acceptance testing and product demo	HEC analysts, university administrators	Months 17 - 19
Technology transfer and hand-over	HEC IT, project team	Months 19 - 20

4D. REFERENCES (cited in 3, 4A, 4B, 4C; not to exceed two pages)

[1] Alzahrani, Bandar, et al. "How ready is higher education for quality 4.0 transformation according to the LNS research framework?" Sustainability 13.9 (2021): 5169.

[2] Conejero, Jose Maria, et al. "Towards the use of Data Engineering, Advanced Visualization techniques and Association Rules to support knowledge discovery for public policies." Expert Systems with Applications 170 (2021): 114509.

[3] Lokaadinugroho, Indrabudhi, Abba Suganda Girsang, and Burhanudin Burhanudin. "Tableau Business Intelligence Using the 9 Steps of Kimball's Data Warehouse and Extract Transform Load Data Integration Process Approach in Higher Education." EMACS Journal 3.1 (2021): 1-11.

[4] Edastama, Primasatria, Amitkumar Dudhat, and Giandari Maulani. "Use of Data Warehouse and Data Mining for Academic Data: A Case Study at a National University." International Journal of Cyber and IT Service Management 1.2 (2021): 206-215.

[5] Abdul Jalil, Nasir, and Mikkay Wong Ei Leen. "Learning Analytics in Higher Education: The Student Expectations of Learning Analytics." 2021 5th International Conference on Education and E-Learning. 2021.

[6] Santoso, Leo Willyanto. "Data warehouse with big data technology for higher education." Procedia Computer Science 124 (2017): 93-99.

[7] Hamad, Faten, et al. "Business intelligence in academic libraries in Jordan: Opportunities and challenges." IFLA Journal 47.1 (2021): 37-50.

[8] Khouri, Selma, Kamel Semassel, and Ladjel Bellatreche. "Managing data warehouse traceability: a life-cycle driven approach." International Conference on Advanced Information Systems Engineering. Springer, Cham, 2015.

[9] Kimball, Ralph, and Margy Ross. The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling, 3rd ed. Wiley, 2013.

[10] Inmon, William H. Building the Data Warehouse, 4th ed. Wiley, 2005.

[11] United Nations General Assembly. Report of the Open Working Group of the General Assembly on Sustainable Development Goals (A/68/970), 2014.

Note: references [1] to [8] are taken from the HEC proposal material supplied by the applicant and should be verified before submission.

5. IMPACT (impact of the proposed prototype on aspects such as transfer of research results into the economy, innovation, import substitution and/or export enhancement, and technology-oriented human-resource development; identify other end-users)

1. Developed countries improve the quality of education through technological means, strive for excellence and expand opportunities based on efficiency and equity; an HEDW brings this capability to Pakistan's higher-education sector.
2. The fundamental role of a data warehouse is to provide data for supporting the decision-making process.
3. Universities and HEC will be able to generate queries and reports on their own, with drill-down by ideology group and use case.
4. Creation of employment and new skills for graduates and post-graduates in data engineering, analytics and governance.
5. HEC can monitor the number of enrolled students in every discipline, institution and province.
6. The warehouse supports a manpower and employment decision-support system by comparing graduate supply with skills demand.
7. It offers a solution to local industrial, societal and economic problems through problem-oriented analytics at university level.
8. University accreditation: the warehouse can produce charts and reports that serve as the representation of institutional activity required by accreditation councils.

End-users: HEC leadership and analysts, provincial higher-education departments, university administrators and quality-enhancement cells, accreditation councils, researchers and, for aggregated indicators, the public.

6. SUSTAINABLE DEVELOPMENT GOALS (SDGs) (how and which of the SDGs will be addressed in this study)

United Nations SDG 4: Quality Education. Quality education for all is fundamental to a peaceful and prosperous world. The goal of this project is to design, develop and implement a data warehouse for HEC Pakistan that enables HEC and government to apply data mining and analytics for better decisions. Key SDG 4 targets addressed:

- **Target 4.3:** Equal access for women and men to affordable, quality technical, vocational and tertiary education. Addressed through enrollment, gender-parity and scholarship-coverage indicators by province and institution (use cases UC01, UC02, UC08).
- **Target 4.4:** More youth and adults with relevant skills for employment. Addressed through graduate-supply and employability analytics (UC04, UC12).
- **Target 4.a:** Build and upgrade education facilities that are safe, inclusive and effective. Addressed through capacity, seat-utilization and institutional-quality indicators (UC02, UC11).
- **Target 4.c:** More qualified teachers. Addressed through faculty-qualification, student-faculty ratio and retention indicators (UC05, UC06).

Supporting goals: SDG 5 (gender equality) and SDG 10 (reduced inequalities) through equity analytics by gender and province; SDG 9 (industry, innovation and infrastructure) and SDG 17 (partnerships) through the data standard and the university-industry-HEC collaboration.

ANNEX A: DIMENSIONAL MODEL DRILL-DOWN

The warehouse uses a star schema with 13 conformed dimensions. Ten are standard higher-education dimensions; three are new analytical dimensions (shown in orange): **Dim_Use_Case**, **Dim_KPI** and **Dim_University_Ideology**. Every dimension has a surrogate key for joins and keeps the business key from the source for traceability.

A.1 Conformed Dimensions around the Core Facts

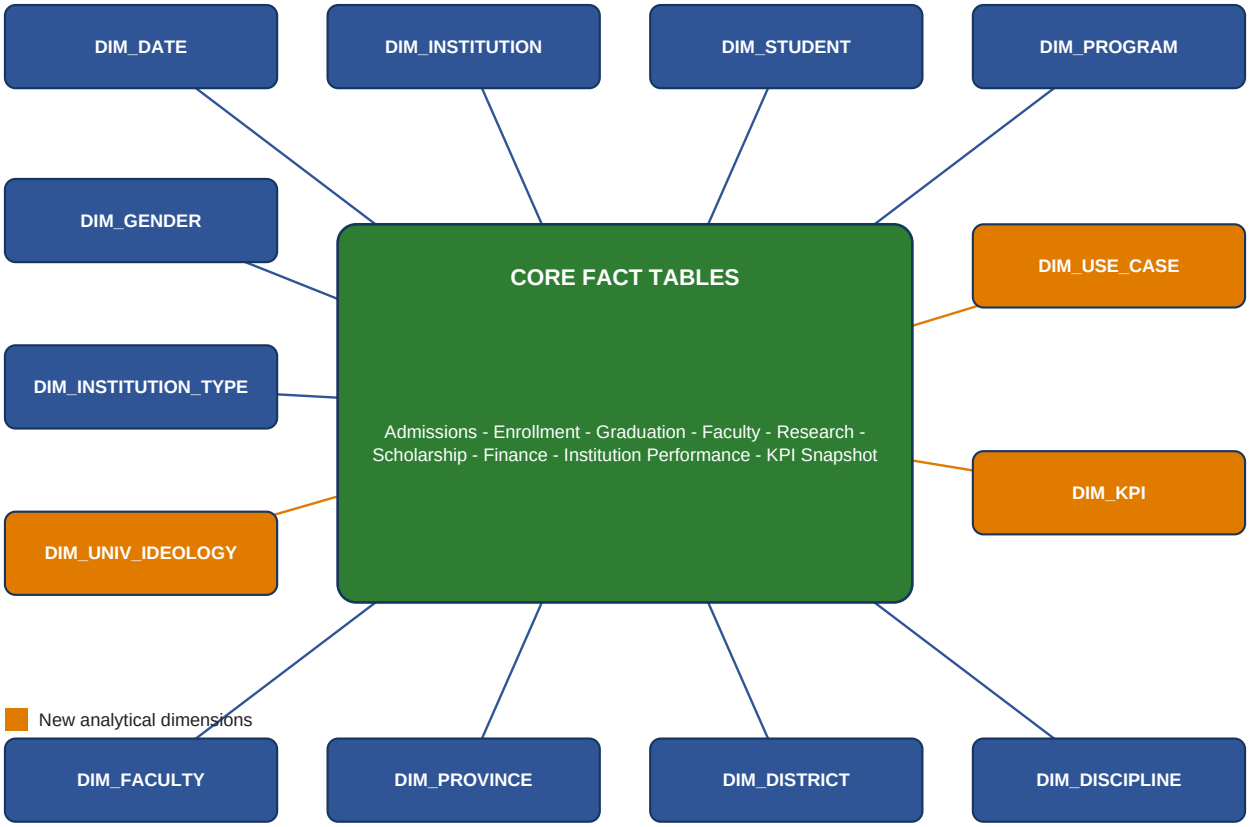


Figure A1: Extended star schema. New dimensions are highlighted.

A.2 Drill-Down Hierarchies

Each hierarchy lets a user start at a national summary and move step by step to the lowest level. Because the University Ideology and Use Case hierarchies are conformed, they can be combined with geography and time in any dashboard, for example *enrollment by ideology group, then by province, then by institution*.

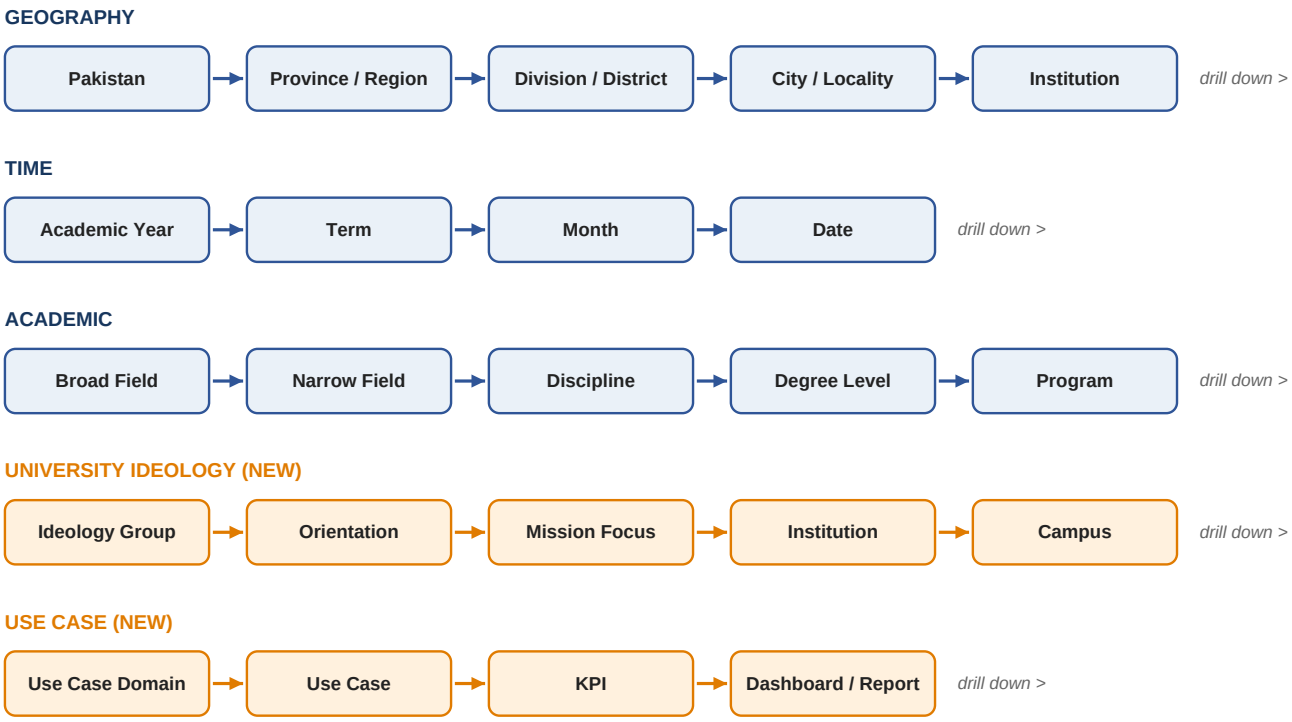


Figure A2: Drill-down paths. New hierarchies are highlighted.

A.3 Dimension Summary

Dimension	Grain	Surrogate Key	SCD	Drill Path
Dim_Date	Day	date_key	Type 0	Academic Year > Term > Month > Date
Dim_Institution	Institution version	institution_key	Type 2	Sector > Type > Institution > Campus
Dim_Student	Student	student_key	Type 1 (+2 status)	Cohort Year > Status > Student
Dim_Program	Program version	program_key	Type 2	Discipline > Level > Program
Dim_Faculty	Faculty version	faculty_key	Type 2	Institution > Discipline > Designation
Dim_Province	Province	province_key	Type 2	Country > Province
Dim_District	District	district_key	Type 2	Province > Division > District > City
Dim_Discipline	Discipline	discipline_key	Type 1	Broad Field > Narrow Field > Discipline
Dim_Gender	Category	gender_key	Type 0	Reporting Group > Category
Dim_Institution_Type	Classification	institution_type_key	Type 1	Sector > Type
Dim_University_Ideology (new)	Classification version	ideology_key	Type 2	Ideology Group > Orientation > Mission Focus
Dim_Use_Case (new)	Use case version	use_case_key	Type 2	Domain > Use Case > KPI > Dashboard
Dim_KPI (new)	KPI version	kpi_key	Type 2	Use Case > KPI > Dashboard

Type	Behavior	Used For
Type 0	Never changes after load.	Date, Gender.
Type 1	Overwrite the old value; no history.	Discipline, Institution Type, most Student attributes.
Type 2	Add a new row with effective_from / effective_to / is_current.	Institution, Program, Faculty, Province, District, Ideology, Use Case, KPI.

A.4 Dimension Usage Matrix

Indicative matrix of which dimensions join to which fact tables. Adm = Admissions, Enr = Enrollment, Grad = Graduation, Fac = Faculty, Res = Research, Sch = Scholarship, Fin = Finance, Perf = Institution Performance, KPI = KPI Snapshot.

Dimension	Adm	Enr	Grad	Fac	Res	Sch	Fin	Perf	KPI
Dim_Date	•	•	•	•	•	•	•	•	•
Dim_Institution	•	•	•	•	•	•	•	•	•
Dim_Student	•	•	•	-	-	•	-	-	-
Dim_Program	•	•	•	-	-	•	-	-	-
Dim_Faculty	-	-	-	•	•	-	-	-	-
Dim_Province	•	•	•	•	•	•	•	•	•
Dim_District	•	•	•	•	•	•	•	•	•
Dim_Discipline	•	•	•	•	•	•	-	-	-
Dim_Gender	•	•	•	•	-	•	-	-	-
Dim_Institution_Type	•	•	•	•	•	•	•	•	•
Dim_University_Ideology	•	•	•	•	•	•	•	•	•
Dim_Use_Case	-	-	-	-	-	-	-	•	•
Dim_KPI	-	-	-	-	-	-	-	-	•

A.5 Dimension Attribute Specifications

Data types are indicative PostgreSQL types. New dimensions are marked (new).

A.5.1 Dim_Date - Calendar and academic time.

Grain	One row per calendar day	**SCD Type**	Type 0 (static)
Surrogate Key	date_key (YYYYMMDD)	**Business Key**	full_date
Drill Path	Academic Year → Term → Month → Date | Calendar Year → Quarter → Month → Date		

Attribute	Data Type	Description / Example
date_key	INT (PK)	Smart key, e.g. 20260915.
full_date	DATE	Actual calendar date.
day_name / day_of_week	VARCHAR / SMALLINT	Monday / 1.
week_of_year	SMALLINT	ISO week number.
month_number / month_name	SMALLINT / VARCHAR	9 / September.
quarter	SMALLINT	Calendar quarter 1–4.
calendar_year	SMALLINT	2026.
academic_year	VARCHAR(9)	2025-2026; central to enrollment and graduation reporting.
academic_term	VARCHAR(10)	Fall / Spring / Summer.
fiscal_year	VARCHAR(9)	Jul–Jun fiscal year, e.g. FY2025-26; used by Fact_Finance.
is_weekend / is_holiday	BOOLEAN	Flags for operational and timeliness analysis.

A.5.2 Dim_Institution - Universities and degree-awarding institutions (DAIs).

Grain	One row per institution per version	**SCD Type**	Type 2 (full history)
Surrogate Key	institution_key	**Business Key**	institution_id + hec_code
Drill Path	Sector → Institution Type → Institution → Campus		

Attribute	Data Type	Description / Example
institution_key	INT (PK)	Surrogate key; changes with each new version.
institution_id	VARCHAR(20)	Stable business identifier; never reused.
hec_code	VARCHAR(20)	Approved HEC/reference code where applicable.
institution_name / short_name	VARCHAR(200 / 30)	Official and abbreviated names (standardized by MDM).
institution_type_key	INT (FK)	Links to Dim_Institution_Type.
ideology_key	INT (FK)	Links to Dim_University_Ideology (mission and orientation).
sector	VARCHAR(10)	Public / Private.
status	VARCHAR(15)	Active / Inactive / Suspended.
established_year	SMALLINT	Year of establishment or charter.
province_key / district_key	INT (FK)	Location of main campus.
campus_type	VARCHAR(15)	Main / Sub-campus / Affiliated.
parent_institution_id	VARCHAR(20)	Self-reference for sub-campuses.
effective_from / effective_to	DATE	Validity window of this version.
is_current	BOOLEAN	True for the latest version.

A.5.3 Dim_Student - Controlled, pseudonymized student attributes only.

Grain	One row per student (pseudonymized)	**SCD Type**	Type 1 (+ Type 2 on status)
Surrogate Key	student_key	**Business Key**	student_id_hash
Drill Path	Admission Year (cohort) → Status → Student		

Attribute	Data Type	Description / Example
student_key	BIGINT (PK)	Surrogate key.
student_id_hash	CHAR(64)	Salted hash of source ID; no direct identifiers stored.
gender_key	INT (FK)	Links to Dim_Gender.
birth_year	SMALLINT	Year only (not full date of birth), for age-band analysis.
domicile_province_key	INT (FK)	Province of domicile (distinct from institution location).
admission_year	SMALLINT	Cohort year for retention and graduation tracking.
admission_category	VARCHAR(20)	Open merit / Self-finance / Reserved quota, as approved.
student_status	VARCHAR(15)	Active / Graduated / Withdrawn / Suspended.
nationality_group	VARCHAR(15)	Domestic / International.

A.5.4 Dim_Program - Degree programs offered by institutions.

Grain	One row per program per institution per version	**SCD Type**	Type 2
Surrogate Key	program_key	**Business Key**	program_id
Drill Path	Discipline → Degree Level → Program		

Attribute	Data Type	Description / Example
program_key	INT (PK)	Surrogate key.
program_id	VARCHAR(30)	Business identifier, unique within institution.
program_title	VARCHAR(200)	e.g. BS Data Science.
degree_level	VARCHAR(15)	Associate / BS / MS-MPhil / PhD.
duration_years	NUMERIC(3,1)	Nominal duration, e.g. 4.0.
total_credit_hours	SMALLINT	Credit hours required for completion.
program_mode	VARCHAR(15)	Regular / Evening / Distance.
discipline_key	INT (FK)	Links to Dim_Discipline.
institution_key	INT (FK)	Offering institution.
accreditation_status	VARCHAR(20)	Accredited / Under review / Not applicable.
effective_from / effective_to / is_current	DATE / DATE / BOOLEAN	Version tracking.

A.5.5 Dim_Faculty - Faculty attributes at approved level of detail.

Grain	One row per faculty member (pseudonymized) per version	**SCD Type**	Type 2
Surrogate Key	faculty_key	**Business Key**	faculty_id_hash
Drill Path	Institution → Discipline → Designation → Faculty		

Attribute	Data Type	Description / Example
faculty_key	BIGINT (PK)	Surrogate key.
faculty_id_hash	CHAR(64)	Salted hash of source ID.
designation	VARCHAR(25)	Lecturer / Assistant / Associate Professor / Professor.
highest_qualification	VARCHAR(15)	PhD / MS-MPhil / Masters / Other.
employment_type	VARCHAR(15)	Permanent / Contract / Visiting.
discipline_key	INT (FK)	Primary discipline.
institution_key	INT (FK)	Current institution.
gender_key	INT (FK)	Links to Dim_Gender.
join_year	SMALLINT	Year of joining the institution.
effective_from / effective_to / is_current	DATE / DATE / BOOLEAN	Tracks promotions and qualification changes.

A.5.6 Dim_Province - Province / region reference data.

Grain	One row per province or administrative region	**SCD Type**	Type 2 (rare changes)
Surrogate Key	province_key	**Business Key**	province_code
Drill Path	Country → Province / Region		

Attribute	Data Type	Description / Example
province_key	INT (PK)	Surrogate key.
province_code	VARCHAR(10)	Official code.
province_name	VARCHAR(60)	Standard name from the reference table.
region_type	VARCHAR(25)	Province / Federal territory / Other authorized region.
country	VARCHAR(30)	Pakistan.
effective_from / effective_to	DATE	Supports changes in official classification without redesign.

A.5.7 Dim_District - District and city hierarchy beneath province.

Grain	One row per district	**SCD Type**	Type 2 (rare changes)
Surrogate Key	district_key	**Business Key**	district_code
Drill Path	Province → Division → District → City / Locality		

Attribute	Data Type	Description / Example
district_key	INT (PK)	Surrogate key.
district_code	VARCHAR(12)	Official code.
district_name	VARCHAR(80)	Standard name.
division_name	VARCHAR(80)	Administrative division, where applicable.
province_key	INT (FK)	Parent province.
city_locality	VARCHAR(80)	Main city/locality used for institution location.
urban_rural_class	VARCHAR(10)	Urban / Rural / Mixed, as officially classified.

A.5.8 Dim_Discipline - Standardized discipline taxonomy.

Grain	One row per discipline	**SCD Type**	Type 1
Surrogate Key	discipline_key	**Business Key**	discipline_code
Drill Path	Broad Field → Narrow Field → Discipline		

Attribute	Data Type	Description / Example
discipline_key	INT (PK)	Surrogate key.
discipline_code	VARCHAR(10)	Taxonomy code.
discipline_name	VARCHAR(120)	e.g. Data Science.
narrow_field	VARCHAR(120)	e.g. Computer Science and Information Systems.
broad_field	VARCHAR(120)	e.g. Engineering and Technology; Natural Sciences; Social Sciences; Business; Arts and Humanities; Health.
taxonomy_version	VARCHAR(10)	Version of the classification (can align with an international standard such as ISCED-F).

A.5.9 Dim_Gender - Controlled categories, only where operationally and legally appropriate.

Grain	One row per approved category	**SCD Type**	Type 0 (static)
Surrogate Key	gender_key	**Business Key**	gender_code
Drill Path	Reporting Group → Category		

Attribute	Data Type	Description / Example
gender_key	INT (PK)	Surrogate key.
gender_code	CHAR(1)	Source code, e.g. M / F / O / U.
gender_label	VARCHAR(20)	Display label.
reporting_group	VARCHAR(20)	Grouping used in published indicators; small cells suppressed.

A.5.10 Dim_Institution_Type - Classification of institutions.

Grain	One row per classification	**SCD Type**	Type 1
Surrogate Key	institution_type_key	**Business Key**	type_code
Drill Path	Sector → Type		

Attribute	Data Type	Description / Example
institution_type_key	INT (PK)	Surrogate key.
type_code	VARCHAR(10)	Approved code.
type_name	VARCHAR(60)	e.g. Federal public university, Provincial public university, Private university, Degree-awarding institute.
sector	VARCHAR(10)	Public / Private.
awarding_status	VARCHAR(25)	University / Degree-awarding institute.
description	VARCHAR(200)	Definition from the approved classification.

A.5.11 Dim_University_Ideology (new) - Institutional mission and orientation classification.

Grain	One row per approved classification combination per version	**SCD Type**	Type 2
Surrogate Key	ideology_key	**Business Key**	ideology_code
Drill Path	Ideology Group > Orientation > Mission Focus > Institution > Campus		

Attribute	Data Type	Description / Example
ideology_key	INT (PK)	Surrogate key.
ideology_code	VARCHAR(12)	Approved code, e.g. IDG-03-02.
ideology_group	VARCHAR(40)	Top level: Research-Intensive / Teaching-Focused / Professional and Technical / Specialized Mission-Based / Founding Philosophy and Charter.
orientation	VARCHAR(60)	Second level, e.g. Engineering and technology; Medical and health sciences; Faith-based (as stated in charter); Women's university; Open and distance learning.
mission_focus	VARCHAR(80)	Narrowest level, e.g. Industry-linked applied research; Regional access; Skills development.
teaching_research_balance	VARCHAR(20)	Research-intensive / Balanced / Teaching-focused.
charter_basis	VARCHAR(25)	Federal act / Provincial act / Private charter / Ordinance.
student_body_focus	VARCHAR(15)	Co-educational / Women only / Men only.
language_of_instruction	VARCHAR(20)	English / Urdu / Bilingual / Other.
delivery_model	VARCHAR(20)	On-campus / Distance / Blended.
secondary_orientation	VARCHAR(60)	Optional second tag; version 1 keeps one primary classification per institution.
classification_basis	VARCHAR(120)	Evidence used: charter clause or validated mission statement.
validated_by / classification_version	VARCHAR(40) / VARCHAR(10)	HEC reviewer role and taxonomy version.
effective_from / effective_to / is_current	DATE / DATE / BOOLEAN	Version tracking when an institution is reclassified.

A.5.12 Dim_Use_Case (new) - Analytical use cases the warehouse serves.

Grain	One row per use case per version	**SCD Type**	Type 2
Surrogate Key	use_case_key	**Business Key**	use_case_id
Drill Path	Use Case Domain > Use Case > KPI > Dashboard / Report		

Attribute	Data Type	Description / Example
use_case_key	INT (PK)	Surrogate key.
use_case_id	VARCHAR(10)	Business ID, e.g. UC03.
use_case_name	VARCHAR(120)	e.g. Graduation and retention tracking.
use_case_domain	VARCHAR(40)	Access and Enrollment / Student Success / Faculty / Research / Equity and Funding / Quality and Accreditation / Planning and Policy / Data Governance / Advanced Analytics.
business_question	VARCHAR(250)	The decision question the use case answers.
primary_user_group	VARCHAR(25)	HEC / Provincial HED / University / Analyst / Public.
decision_level	VARCHAR(15)	National / Provincial / Institutional.
analytics_type	VARCHAR(15)	Descriptive / Diagnostic / Predictive / Prescriptive.
priority_phase	VARCHAR(10)	Pilot / Phase 2 / Future.
refresh_frequency	VARCHAR(15)	Monthly / Semester / Annual.
owner_role / status	VARCHAR(40) / VARCHAR(12)	Business owner role; Draft / Approved / Retired.
effective_from / effective_to / is_current	DATE / DATE / BOOLEAN	Version tracking.

A.5.13 Dim_KPI (new) - Governed KPI definitions linked to use cases.

Grain	One row per KPI per version	**SCD Type**	Type 2
Surrogate Key	kpi_key	**Business Key**	kpi_code
Drill Path	Use Case > KPI > Dashboard / Report		

Attribute	Data Type	Description / Example
kpi_key	INT (PK)	Surrogate key.
kpi_code	VARCHAR(10)	Business code, e.g. K06.
kpi_name	VARCHAR(120)	e.g. Graduation rate.
kpi_definition	VARCHAR(300)	Plain-language formula with numerator and denominator.
unit	VARCHAR(15)	Count / Percent / Ratio / Rupees / Index.
polarity	VARCHAR(15)	Higher-is-better / Lower-is-better / Neutral.
use_case_key	INT (FK)	Links to Dim_Use_Case.
source_fact / calc_grain	VARCHAR(40)	Fact table used; grain of calculation, e.g. Institution x Academic Year.
target_value	NUMERIC(14,2)	Policy target where defined; blank otherwise.
dashboard_name	VARCHAR(80)	Dashboard or report showing the KPI.
approval_status	VARCHAR(12)	Draft / Approved / Retired.
effective_from / effective_to / is_current	DATE / DATE / BOOLEAN	Version tracking.

ANNEX B: USE CASE DIMENSION - DRILL-DOWN

The Use Case dimension records *why* a number is needed. Each use case belongs to a domain, has a primary user and decision level, and owns a set of KPIs (Dim_KPI). KPI values are stored in **Fact_KPI_Snapshot**, which carries the keys of date, institution, province, ideology, use case and KPI. This lets any KPI be drilled by use case, institution, province or ideology group.

B.1 Use Case Catalogue

ID	Domain	Use case and business question	Primary users	Phase
UC01	Access and Enrollment	Enrollment monitoring: how many students are enrolled by year, province, institution, program and discipline?	HEC, Provincial HED	Pilot
UC02	Access and Enrollment	Admissions and intake utilization: how are applications, admissions and seat utilization changing?	HEC, University	Pilot
UC03	Student Success	Graduation and retention tracking: what share of each cohort completes, continues or withdraws?	HEC, University	Pilot
UC04	Student Success	Graduate employability: where do alumni work and how does it vary by discipline?	HEC, Analyst	Phase 2
UC05	Faculty	Faculty availability and qualification: student-faculty ratio and share of PhD faculty.	HEC, University	Pilot
UC06	Faculty	Faculty retention: attrition and promotion patterns by designation.	University, HEC	Phase 2
UC07	Research	Research output and funding: where is research activity concentrated and how is it changing?	HEC, Analyst	Phase 2
UC08	Equity and Funding	Scholarship distribution and equity: who receives support, by gender and province?	HEC, Provincial HED	Pilot
UC09	Equity and Funding	Funding indicators: how do approved funding figures vary by institution type and province?	HEC	Phase 2
UC10	Quality and Accreditation	Accreditation readiness: which programs and institutions are accredited or due for review?	HEC, University	Pilot
UC11	Quality and Accreditation	Institutional performance comparison: how do institutions compare within the same ideology group?	HEC	Pilot
UC12	Planning and Policy	Manpower planning: how does graduate supply by discipline compare with demand?	HEC, Provincial HED	Phase 2
UC13	Planning and Policy	Program gap and duplication: which disciplines are over- or under-served by province?	HEC, Provincial HED	Phase 2
UC14	Data Governance	Submission and data-quality monitoring: who submitted on time, and how complete and valid is the data?	HEC data stewards	Pilot
UC15	Advanced Analytics	Early warning: which cohorts show higher dropout risk (aggregate level)?	University, Analyst	Future
UC16	Advanced Analytics	Enrollment forecasting: expected enrollment by discipline and province over the next three years.	HEC, Planning	Future

B.2 KPI Catalogue (indicative)

Definitions are proposals and will be finalized in the Business Requirements Specification.

KPI	Name	Definition	UC	Source fact
K01	Total enrollment	Count of enrolled students in the period.	UC01	Enrollment
K02	Enrollment growth	Change in enrollment versus the previous academic year, in percent.	UC01	Enrollment
K03	Gender parity index	Female enrollment divided by male enrollment.	UC01	Enrollment
K04	Seat utilization	Admitted students divided by approved intake, in percent.	UC02	Admissions
K05	Application-to-admission ratio	Applications received per admission offered.	UC02	Admissions
K06	Graduation rate	Graduates of a cohort within normal duration divided by cohort entrants.	UC03	Graduation
K07	Retention rate	Cohort students continuing into the next year divided by cohort size.	UC03	Enrollment
K08	Withdrawal rate	Cohort students who withdrew divided by cohort size.	UC03	Enrollment

KPI	Name	Definition	UC	Source fact
K09	Graduate employment rate	Alumni employed within a defined period divided by alumni surveyed.	UC04	Alumni outcome (Phase 2)
K10	Student-faculty ratio	Enrolled students divided by full-time-equivalent faculty.	UC05	Enrollment, Faculty
K11	Faculty with PhD	Faculty holding a PhD divided by total faculty, in percent.	UC05	Faculty
K12	Faculty attrition rate	Faculty who left divided by faculty at start of year.	UC06	Faculty
K13	Research outputs per faculty	Approved research outputs divided by faculty count.	UC07	Research
K14	Research funding per faculty	Research funding received divided by faculty count (Rs).	UC07	Research
K15	Scholarship coverage	Scholarship holders divided by enrolled students, in percent.	UC08	Scholarship
K16	Scholarship share by gender and province	Share of scholarship awards by gender and province.	UC08	Scholarship
K17	Funding per student	Approved funding divided by enrolled students (Rs).	UC09	Finance
K18	Accredited programs	Accredited programs divided by total programs, in percent.	UC10	Institution Performance
K19	Institutional quality index	Composite of approved quality indicators (weights set by HEC).	UC11	Institution Performance
K20	Supply-demand index	Graduates by discipline relative to estimated demand.	UC12	Graduation (+ external demand data)
K21	Programs per discipline per province	Count of programs offered by discipline and province.	UC13	Enrollment
K22	On-time submission	Submissions received by the deadline divided by expected, in percent.	UC14	Load metadata
K23	Data completeness	Required fields populated divided by required fields, in percent.	UC14	DQ results
K24	Data validity pass rate	Records passing validation rules divided by records loaded.	UC14	DQ results
K25	Early-warning risk flags	Cohort-level count of risk flags (aggregate, no individual labels).	UC15	Enrollment, Graduation
K26	Enrollment forecast	Forecast enrollment for the next three years.	UC16	Enrollment

B.3 Fact_KPI_Snapshot

Column	Data Type	Description
date_key / institution_key / province_key	INT (FK)	When and where the KPI applies.
ideology_key	INT (FK)	Copied from the institution's current classification to speed up drill-down.
use_case_key / kpi_key	INT (FK)	Which decision and which KPI the value belongs to.
kpi_value	NUMERIC(18,4)	Calculated value.
numerator / denominator	NUMERIC(18,4)	Stored so values can be re-aggregated correctly at any level.
period_type	VARCHAR(10)	Semester / Annual / Monthly.
load_batch_id / dq_status	BIGINT / VARCHAR(10)	Lineage to the load and its data-quality result.

B.4 Example Drill-Down: UC03 Graduation and Retention

Level	Example view	Example question
1 National	Graduation rate (K06) for Pakistan	Is the national graduation rate improving year on year?
2 Province	K06 by province	Which provinces are below the national figure?
3 Ideology group	K06 by ideology group within the province	Do teaching-focused and research-intensive institutions differ?
4 Institution	K06 by institution within the group	Which institutions are outliers against their peer group?
5 Program	K06 by program and cohort year	Which programs drive the gap?
6 Numerator / denominator	Cohort entrants and graduates	Is the change due to intake size or completions?

ANNEX C: UNIVERSITY IDEOLOGY DIMENSION - DRILL-DOWN

C.1 Purpose and Interpretation

The University Ideology dimension classifies each institution by its **founding philosophy, mission and academic orientation**, so that institutions with similar purposes are compared with each other and differences in outcomes can be explained. For example, a research-intensive institution and a regional teaching institution should not be judged on identical expectations.

Design rules: (1) classification is based on documented evidence such as the institution's charter or ordinance and a validated mission statement, never on subjective judgment; (2) HEC validates each classification and the taxonomy is versioned (Type 2); (3) classification describes the *institution* only, not individual students or staff; (4) version 1 keeps one primary classification per institution, with an optional secondary tag; (5) the taxonomy is reference data and can change without redesigning fact tables.

C.2 Classification Taxonomy (proposed)

Ideology Group	Orientation (level 2)	Mission Focus examples (level 3)
Research-Intensive	Research university; Graduate-research focused	Basic sciences; Applied and engineering research; Multidisciplinary research
Teaching-Focused	Undergraduate teaching; Regional teaching	General education; Regional access and community service
Professional and Technical	Engineering and technology; Business and management; Vocational and skills	Industry-linked programs; Skills development; Professional licensing pathways
Specialized Mission-Based	Medical and health sciences; Agriculture and veterinary; Arts and design; Law; Education; Defence-services-linked	Sector-specific training and research
Founding Philosophy and Charter	Faith-based (as stated in charter); Women's university; Provincial or regional development; Open and distance learning	As declared in the charter and validated by HEC

C.3 Drill-Down Path with Examples

Level	Example member	Example analysis
1 All institutions	Pakistan	Total enrollment and graduation by year.
2 Ideology group	Professional and Technical	How does enrollment growth compare with Research-Intensive institutions?
3 Orientation	Engineering and technology	What is the student-faculty ratio and share of PhD faculty?
4 Mission focus	Industry-linked programs	How do graduation and employability compare with other focuses?
5 Institution	Institution A	Where does Institution A stand against its peer group?
6 Campus / Program	Main campus, BS program	Which programs drive the institution's result?

C.4 Questions the Dimension Enables

- Compare graduation, retention and faculty quality across ideology groups instead of across all institutions at once.
- Check whether scholarship coverage and gender parity differ by orientation or by student-body focus.
- Benchmark an institution against its true peer group for accreditation and quality reviews (UC10, UC11).
- Plan program expansion by region and mission (UC13), for example technical programs in under-served districts.
- Slice any KPI in Fact_KPI_Snapshot by ideology group, then combine with province and time.

ANNEX D: PRODUCT REQUIREMENTS, RISKS AND NEXT STEP

D.1 Functional Requirements

ID	Requirement	Priority
FR-01	Accept institutional data through Excel/CSV templates, database connectors and secure file transfer.	Must
FR-02	Keep an unchanged raw copy of every submission for audit.	Must
FR-03	Validate schema, data types, required fields and business rules; route failures to an error table with reasons.	Must
FR-04	Apply a quality gate (completeness, accuracy, consistency, uniqueness, validity, timeliness) and publish a quality scorecard per load.	Must
FR-05	Maintain master data for institutions, programs, disciplines, geography and ideology classification with stable identifiers.	Must
FR-06	Load conformed dimensions (with Type 0/1/2 handling) and fact tables incrementally.	Must
FR-07	Provide data marts for enrollment, admissions, graduation, faculty, research, scholarship, finance and quality.	Must
FR-08	Store KPI definitions (Dim_KPI) and KPI values (Fact_KPI_Snapshot) by use case.	Must
FR-09	Support drill-down by geography, time, discipline, ideology group and use case in Power BI.	Must
FR-10	Provide national, provincial and institutional dashboards with role-appropriate access.	Must
FR-11	Record lineage and metadata: source, load batch, refresh time and business definition.	Should
FR-12	Expose approved aggregated indicators to public users.	Could
FR-13	Provide a governed API layer for future applications.	Could
FR-14	Provide forecasting and early-warning analytics on governed warehouse data.	Could

D.2 Non-Functional Requirements

ID	Non-functional requirement	Priority
NFR-01	Role-based and row-level access control; encryption in transit and at rest.	Must
NFR-02	Direct identifiers are hashed or excluded; small-cell suppression in published indicators.	Must
NFR-03	Audit logging of data access and administrative changes.	Must
NFR-04	Reproducible, scheduled and monitored pipeline runs with failure alerts.	Should
NFR-05	Backup, restore testing and disaster-recovery procedure.	Should
NFR-06	Scalable design: partitioning and indexing for growing fact tables; layers deployable separately.	Should
NFR-07	Documentation: data dictionary, data standard, runbooks and user guides.	Must
NFR-08	Cloud-compatible architecture without vendor lock-in for the core model.	Could

D.3 Key Risks and Mitigation

Risk	Mitigation
Institutions submit late or incomplete data.	Clear submission calendar, templates, timeliness KPI (K22) and steward follow-up.
Privacy and legal concerns about student and faculty data.	Minimize and pseudonymize; use synthetic or approved data in the pilot; follow applicable data-protection rules.
Disagreement over institutional classification.	Evidence-based rules, HEC validation, versioned taxonomy and appeal process.
Different definitions of the same KPI across institutions.	Governed KPI catalogue (Dim_KPI) approved by HEC data owners.
Scope creep beyond the pilot.	Use cases tagged Pilot / Phase 2 / Future; change control through the steering committee.
Limited resources for development and operations.	Phased delivery, open-source core stack, and a defined hand-over to HEC.

D.4 Immediate Next Step

Do not start ETL coding yet. The next deliverable is the **Business Requirements Specification (BRS)**: stakeholders, business questions, final KPI definitions, data ownership and acceptance criteria, followed by the physical PostgreSQL model of the 13 dimensions and 9 fact tables.